

Operational Requirements Document (ORD) & System Requirements Specification (SRS)

1. Document Overview & Project Purpose

Project Name: SSGI SecureDoc – Web-Based Read-Only Digital Library System

Project Purpose:

The SSGI SecureDoc Digital Library System is a secure web-based platform that provides centralized access to research publications, geospatial training materials, reports, and academic documents. The system protects institutional intellectual property through a secure read-only, zero-download architecture while allowing authorized users to access resources online.

Target Users

- Registered Trainees/Interns
- Staff Members
- Librarians
- System Administrators

2. Operational Workflows & User Profiles

Users: Search, browse, and securely read digital resources online, receive notifications, and receive assistance through the AI chatbot.

Librarians: Upload and organize documents, manage metadata and categories, archive resources, and publish announcements.

System Administrators: Manage users and roles, monitor system activity, generate reports, maintain security, and oversee overall system operations.

3. Functional System Requirements

Function	Description
User Authentication	Secure login, password recovery, and role-based access control (RBAC).
Search & Discovery	Search documents by title, author, category, keywords, and serial number.
Document Viewing	Secure read-only viewing with dynamic watermark protection.

Function	Description
Library Management	Upload, categorize, update, and archive digital resources with metadata management.
User Management	Create and manage users, roles, permissions, and activity logs.
Notifications	Send announcements and notifications to users based on their roles.
AI Chatbot	Assist users with document searches and provide basic information and guidance.

4 Non-Functional Requirements

Performance: The system shall return search results quickly, and documents shall begin streaming within two seconds under normal network conditions.

Security: Passwords shall be encrypted, document storage shall be protected, and all user activities shall be recorded for auditing.

Usability: The system shall provide a responsive and user-friendly interface compatible with modern web browsers.

Maintainability: The system shall support regular maintenance, database backups, and future software updates with minimal disruption.

5. Verification & Acceptance Criteria

- Users shall successfully log in, search, and view digital documents according to their assigned roles.
- Librarians shall upload, organize, and manage library resources successfully.
- Administrators shall manage users, permissions, and system activities.
- Unauthorized users shall not access protected resources or download restricted documents.
- Search results and document loading shall meet the required performance standards.

6. Conclusion

The SSGI SecureDoc Digital Library System provides a secure, reliable, and efficient solution for managing institutional digital resources. By integrating secure authentication, role-based access control, protected online document viewing, AI-assisted support, and comprehensive activity logging, the system enhances knowledge sharing while protecting valuable institutional information. These operational and system requirements provide a solid foundation for the successful development, deployment, and long-term maintenance of the Digital Library System.